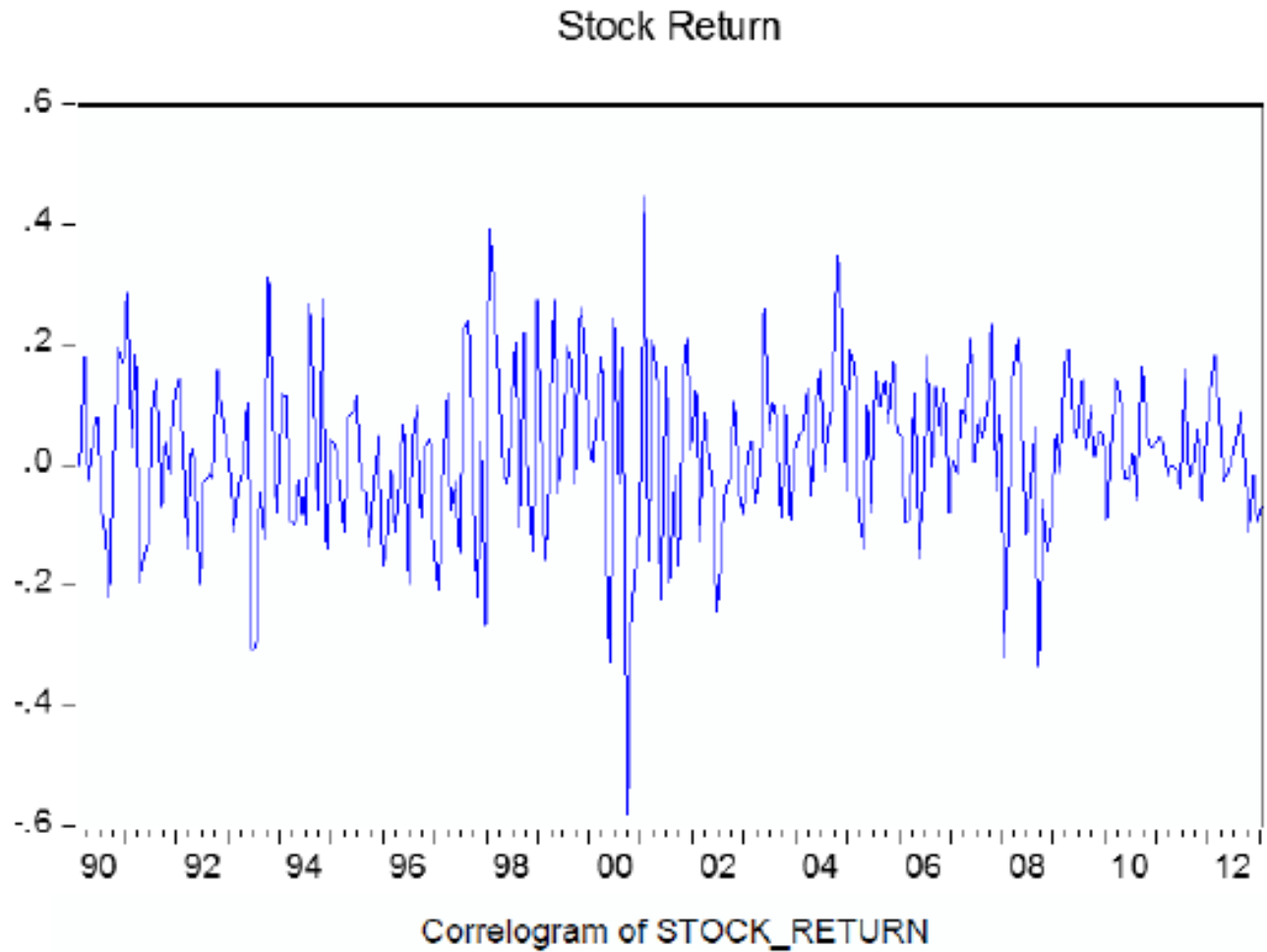

Time Series Econometrics

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Semester 1 2016

1. Below is a plot of IBM monthly stock returns from Feb 1991-Jan 2013 with corresponding correlogram up to lag 5, (T=276).



Date: 10/21/13 Time: 18:45						
Sample: 1990M02 2013M01						
Included observations: 276						
Autocorrelation	Partial Correlation	AC	PAC	Q-Stat	Prob	
		1	0.064	0.064	1.1294	0.288
		2	0.023	0.020	1.2841	0.526
		3	-0.018	-0.020	1.3702	0.713
		4	-0.115	-0.114	5.1043	0.277
		5	0.068	0.085	6.4190	0.268

- (a) Comment on the properties of Stock Returns from the graph.
- (b) Test at the 5% significance level of the hypothesis that the autocorrelation at lag 4 is equal to zero.
- (c) Perform a test at the 10% significance level the hypothesis that the first 5 autocorrelations are jointly equal to zero.
- (d) From the evidence in part (a)-(c) comment on the likely form of the underlying process of IBM stock returns.

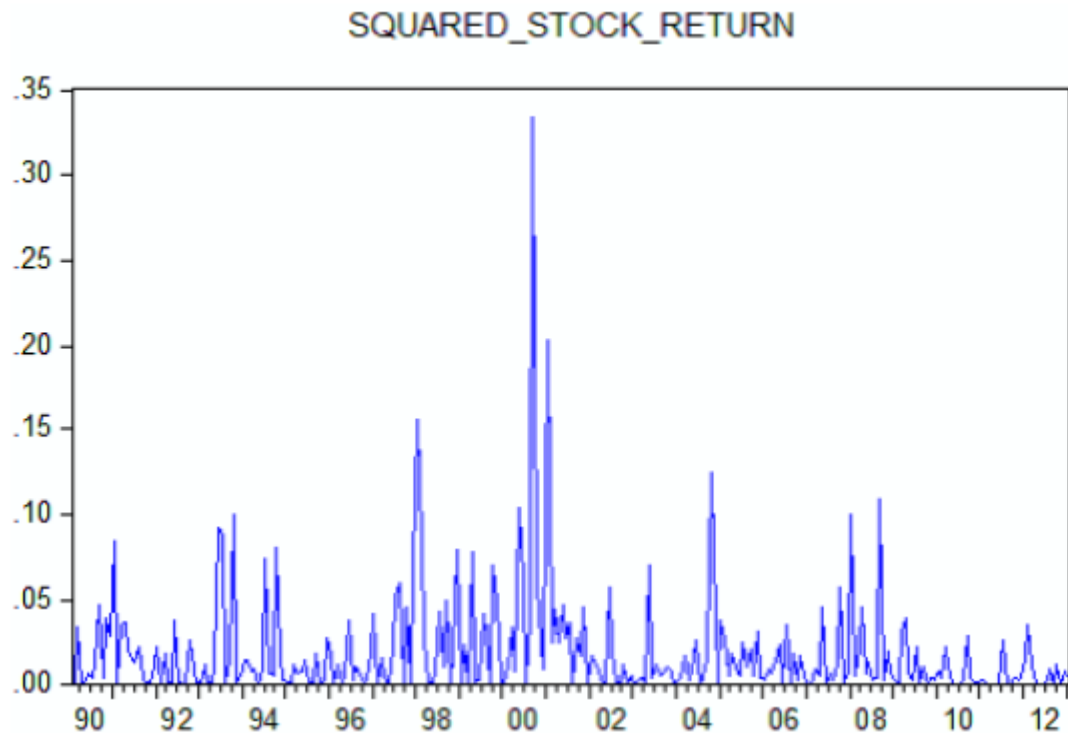
Below are the estimated output of the MA processes- $Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1}$ and $Y_t = \varepsilon_t + \theta_2 \varepsilon_{t-4}$.

Dependent Variable: STOCK_RETURN Method: Least Squares Date: 10/21/13 Time: 18:42 Sample: 1990M02 2013M01 Included observations: 276 Convergence achieved after 5 iterations MA Backcast: 1990M01				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.024804	0.008720	2.844522	0.0048
MA(1)	0.061005	0.060330	1.011191	0.3128
R-squared	0.003882	Mean dependent var	0.024828	
Adjusted R-squared	0.000246	S.D. dependent var	0.136582	
S.E. of regression	0.136565	Akaike info criterion	-1.136805	
Sum squared resid	5.110136	Schwarz criterion	-1.110570	

Dependent Variable: STOCK_RETURN Method: Least Squares Date: 10/21/13 Time: 18:43 Sample: 1990M02 2013M01 Included observations: 276 Convergence achieved after 5 iterations MA Backcast: 1989M10 1990M01				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.024953	0.007181	3.474880	0.0006
MA(4)	-0.123749	0.060155	-2.057161	0.0406
R-squared	0.014377	Mean dependent var	0.024828	
Adjusted R-squared	0.010779	S.D. dependent var	0.136582	
S.E. of regression	0.135844	Akaike info criterion	-1.147397	
Sum squared resid	5.056297	Schwarz criterion	-1.121162	

- (e) Using this estimation output discuss the properties of the true underlying process, performing formal tests. Does this correspond with the discussion in (d)?

2. We now perform a similar analysis as in parts 1 (a)-(e) but for the square of IBM monthly stock returns over the same period.



Date: 10/21/13 Time: 18:31 Sample: 1990M02 2013M01 Included observations: 276						
Autocorrelation	Partial Correlation		AC	PAC	Q-Stat	Prob
		1	0.144	0.144	5.7786	0.016
		2	0.014	-0.007	5.8360	0.054
		3	0.131	0.133	10.664	0.014
		4	0.341	0.315	43.376	0.000
		5	0.043	-0.043	43.894	0.000

- (a) Comment on the properties of squared stock returns from the graph.
- (b) Test at the 5% significance level the hypothesis that the autocorrelation at lag 4 is equal to zero.
- (c) Perform a test at the 10% significance level the hypothesis that the first 5 autocorrelations are jointly equal to zero
- (d) From the graph and information in parts (f)-(h) comment on the properties of the theoretical process of

the square of IBM stock returns?

Dependent Variable: SQUARED_STOCK_RETURN Method: Least Squares Date: 10/21/13 Time: 18:24 Sample: 1990M02 2013M01 Included observations: 276 Convergence achieved after 4 iterations MA Backcast: 1990M01				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.019186	0.002261	8.487168	0.0000
MA(1)	0.147673	0.059764	2.470945	0.0141
R-squared	0.021141	Mean dependent var		0.019204
Adjusted R-squared	0.017568	S.D. dependent var		0.033029
S.E. of regression	0.032738	Akaike info criterion		-3.993352
Sum squared resid	0.293663	Schwarz criterion		-3.967117

Dependent Variable: SQUARED_STOCK_RETURN Method: Least Squares Date: 10/21/13 Time: 18:29 Sample: 1990M02 2013M01 Included observations: 276 Convergence achieved after 8 iterations MA Backcast: 1989M10 1990M01				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.019133	0.002730	7.009356	0.0000
MA(1)	0.103565	0.057614	1.797566	0.0734
MA(2)	-0.044011	0.057688	-0.762914	0.4462
MA(3)	0.087327	0.057746	1.512267	0.1316
MA(4)	0.315124	0.057642	5.466928	0.0000
R-squared	0.125990	Mean dependent var		0.019204
Adjusted R-squared	0.113090	S.D. dependent var		0.033029
S.E. of regression	0.031106	Akaike info criterion		-4.084909
Sum squared resid	0.262208	Schwarz criterion		-4.019322

- (e) Using the estimated output from an MA(1) and MA(4) below discuss the properties of the underlying process, providing formal argument for you answer . Does this correspond with the intuition from part (d)?

3. Suppose a researcher believes the true process is AR(1), i.e

$$Y_t = \mu + \phi_1 Y_{t-1} + \varepsilon_t \quad (1)$$

- (a) If $Cov(\varepsilon_t, \varepsilon_{t-1}) \neq 0$ will OLS applied in (1) provide a consistent estimator of ϕ_1 ? Discuss. [No formal proof needed.]
- (b) Suppose ε_t is a stationary mean zero ARMA(1,1), i.e

$$\varepsilon_t = \gamma_1 \varepsilon_{t-1} + v_t + \kappa_1 v_{t-1} \quad (2)$$

for some coefficients γ_1, κ_1 where $v_t \sim \text{WN}(\sigma^2)$. Given (1) and (2) derive the true form of the process Y_t .